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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Third Semester B.Tech Degree Examination December 2020 (2019 Scheme)

Course Code: HUT200
Course Name: PROFESSIONAL ETHICS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks Marks

1	Why sharing and caring are important for a professional?	(3)
Ans	Caring is expressing concern about others, their feelings and well being. Caring shouldn't be limited to one's family and friends. Caring should be given to neighbors, colleagues, with whom we deal with in our daily life, etc. Caring for environment and Nature should be there as it is the necessity of time. Sharing of knowledge, facilities, goods, experiences, etc lead to the growth of society. The act of sharing should come voluntarily and without compulsion. It leads to peaceful living.	
2	Define work Ethics.	(3)
	Work ethics can be defined as a set of standards of behaviour or codes of conduct based on a set of values, in the workplace. Medical ethics, engineering ethics are examples. A strong set of work ethics promotes the well being of employees, organisational effectiveness and advancement of society. The basic elements of a well formed code of work ethics in an oraganisation are given below ● Integrity and Loyalty ● Professionalism ● Respect and Care ● Cooperation ● Fairness ● Trustworthiness	
3	What are the different types of enquiries in solving ethical problems?	(3

		)
	Enquiries in engineering ethics fall under three categories namely, **1. Normative Enquiries** Normative questions deal with what ought to be done. It is about identifying the values and practices that are morally right and ought to guide the decision makers. Example of normative question: - what are the values that should guide a person working in genetic engineering? **2. Conceptual Enquiries** These enquiries throw light into the meaning of concepts, Values, principles and issues encountered in engineering ethics. Example: - what differentiates a bribe from an acceptable gift? **3. Factual Enquiries** : These enquiries provide information regarding the moral practices of an engineer, an organisation or even a society without judging their moral rightness. This can also apply to questions regarding the facts about a moral problem related to engineering. Example: - What caused the first Tesla Model S (self drive car) crash in May2016?	
4	What is moral autonomy?	(3)
	Moral autonomy is the capacity to think rationally and decide what is right instead of simply following a set of rules. It is about making decisions based on moral concern by reasoning rather than blindly following others. **Moral autonomy is about the following,** ● Moral concern ● Going through different perspectives of an ethical issue ● Application of reasoning ● Making decisions without the influence of others ● Acting boldly on the decision taken	
5	List out the models of professional roles.	(3)
	Engineers often identify themselves with some of the roles given below. **1. Saviour :** Engineer assumes the role of someone who saves the society from all its evils. He may use technology to save them from poverty and ill health. By using appropriate and advanced technology, efficiency and effectiveness can be ensured in all activities related to social planning which intern results in material prosperity and happiness. **2. Guardian** : An engineer can play a crucial role in determining the needs of the society.	

	He can guide and regulate the technological advancement keeping in mind the best interest and welfare of the common man. **3. Bureaucratic Servant** : Here the role of engineer is confined to the development of engineering solutions to the problems handed to them by the organisation. The management decides what to develop and the engineer assigned the technical side of the problem. His duty is to come up with concrete engineering solutions.	
6	Define plagiarism.	(3)
	Plagiarism is presenting someone else's work or ideas as your own, with or without their consent, by incorporating it into your work without full acknowledgement. All published and unpublished material, whether in manuscript, printed or electronic form, is covered under this definition. Plagiarism may be intentional or reckless, or unintentional. Under the regulations for examinations, intentional or reckless plagiarism is a disciplinary offence. The necessity to acknowledge others' work or ideas applies not only to text, but also to other media, such as computer code, illustrations, graphs etc. It applies equally to published text and data drawn from books and journals, and to unpublished text and data, whether from lectures, theses or other students' essays. You must also attribute text, data, or other resources downloaded from websites.	
7	What is the significance of intellectual property rights?	(3)
	▪ Intellectual property rights (IPR) are the rights given to persons over the creations of their minds: inventions, literary and artistic works, and symbols, names and images used in commerce. They usually give the creator an exclusive right over the use of his/her creation for a certain period of time. ▪ These rights are outlined in **Article 27 of the Universal Declaration of Human Rights,** which provides for the right to benefit from the protection of moral and material interests resulting from authorship of scientific, literary or artistic productions. ▪ The importance of intellectual property was first recognized in the **Paris Convention for the Protection of Industrial Property (1883)** and the **Berne Convention for the Protection of Literary and Artistic**	

Works (1886). Both treaties are administered by the **World Intellectual Property Organization (WIPO)**.

8 What is the difference between a bribe and a gift? (3)

The conflict arises when accepting large gifts from the suppliers. Bribe is different from a gift. The following table shows a comparison of the nature of bribe and gift.

Tests	*Bribe*	*Gift*
1. Timing	Given before	Given after
2. Cost of item	Large amount	Small amount, articles of daily use
3. Quality of product	Poor	Good/High
4. Giver is a friend	Yes	No
5. Transparency	Made in secret	Made in open
6. Motive	Expect undue favor	Expect a favor or thanking for the favor
7. Consequence on organization's goodwill	Damaging the goodwill and reputation	No damage is involved

9 What is business ethics? (3)

Sarbans-Oxley Act, 2002 (U.S.A.) and New York Stock Exchange listing standards have made many corporate organizations ethics conscious. The organizations are to disclose codes of business conduct and ethics for the organizations. For example, Texas Instruments a major MNC, has declared that "Ethical reputation is our vital asset. Upon applying values each of the employee can say, TI is a good company, and one reason is that I am part of it". Three major values such as Integrity, Innovation and Commitment, have been elaborated in the form of 28 ethics statements (as pledges to keep) and 17 codes of business conduct have been presented in their documents.

A quick ethics test suggested by TI for all of its employees, without exception, will sufficiently explain their commitment:

1. Is the action LEGAL?
2. Does it comply with our VALUES?

	3. If you do it, will you feel BAD? 4. How will it look in the NEWSPAPER? 5. If you know it is WRONG, don't do it. 6. If you are not sure, ASK 7. Keep asking until you get an answer.	
10	Differentiate between patent and trade secret.	(3)
	Patent is a contract between the individual (inventor) and the society (all others). Patents protect legally the specific products from being manufactured or sold by others, without permission of the patent holder. Patent holder has the legally-protected monopoly power as one's own property. The validity is 20 years from the date filing the application for the patent. It is a territorial right and needs registration. The Patent (Amendment) Act 2002 guarantees such provisions. Patent is given to a product or a process, provided it is entirely new, involving an inventive method and suitable for industrial application. While applying for a patent, it is essential to submit the documents in detail regarding the problem addressed, its solution, extent of novelty or innovation, typical applications, particulars of the inventor, and the resources utilized. Inventions are patentable and the discoveries are not. **4. Trade Secret** A trade secret is the information which is kept confidential as a secret. This information is not accessed by the any other (competitor) than the owner and this gives a commercial advantage over the competitors. The trade secrets are not registered but only kept confidential. These are given limited legal protection, against abuse by the employee or contractor, by keeping confidentiality and trust. The trade secrets may be formulae, or methods, or programs, or processes or test results or data collected, analyzed, and synthesized. These are related to designs, technical processes, plant facilities, list of suppliers or customers etc. This information should not be disclosed or used by any other person.	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11(a)	With the help of examples, distinguish between 'morality' and 'ethics'.	(7)
	Morality is concerned with principles and practices of morals such as: (a) What ought or ought not to be done in a given situation? (b) What is right or wrong about the handling of a situation? and	

	(c) What is good or bad about the people, policies, and ideals involved?	

Morality	***Ethics***
1. More general and prescriptive based on customs and traditions.	1. Specific and descriptive. It is a critical reflection on morals.
2. More concerned with the results of wrong action, when done.	2. More concerned with the results of a right action, when not done.
3. Thrust is on judgment and punishment, in the name of God or by laws.	3. Thrust is on influence, education, training through codes, guidelines, and correction.
4. In case of conflict between the two, morality is given top priority, because the damage is more. It is more common and basic.	4. Less serious, hence second priority only. Less common. But relevant today, because of complex interactions in the modern society.
5. Example: Character flaw, corruption, extortion, and crime.	5. Example: Notions or beliefs about manners, tastes, customs, and towards laws.

(b)	Explain the different aspects of academic integrity.	(7)
	Integrity is defined as the unity of thought, word and deed (honesty) and open mindedness. It includes the capacity to communicate the factual information so that others can make well-informed decisions. It yields the person's 'peace of mind', and hence adds strength and consistency in character, decisions, and actions. This paves way to one's success. It is one of the self-direction virtues. It enthuse people not only to execute a job well but to achieve excellence in performance. It helps them to own the responsibility and earn self-respect and recognition by doing the job. **Academic Integrity :** Any academic endeavour must be pure and away from plagiarism. People in academic community should stick on to truthful information. A student who copies an assignment, a researcher who fabricate data, a writer who doesn't acknowledge his sources, etc lack academic integrity.	
12(a)	Explain the different types of human values.	(7)
	The five core human values are: (1) Right conduct, (2) Peace , (3) Truth, (4) Love, and (5) Nonviolence. 1. Values related to RIGHT CONDUCT are:	

	• SELF-HELP SKILLS: Care of possessions, diet, hygiene, modesty, posture, self reliance, and tidy appearance • SOCIAL SKILLS: Good behavior, good manners, good relationships, helpfulness, No wastage, and good environment, and • ETHICAL SKILLS: Code of conduct, courage, dependability, duty, efficiency, ingenuity, initiative, perseverance, punctuality, resourcefulness, respect for all, and responsibility 2. Values related to PEACE are: Attention, calmness, concentration, contentment, dignity, discipline, equality, equanimity, faithfulness, focus, gratitude, happiness, harmony, humility, inner silence, optimism, patience, reflection, satisfaction, self-acceptance, self-confidence, self-control, self-discipline, self-esteem, self-respect, sense control, tolerance, and understanding 3. Values related to TRUTH are: Accuracy, curiosity, discernment, fairness, fearlessness, honesty, integrity (unity of thought, word, and deed), intuition, justice, optimism, purity, quest for knowledge, reason, self-analysis, sincerity, sprit of enquiry, synthesis, trust, truthfulness, and determination. 4. Values related to LOVE are: Acceptance, affection, care, compassion, consideration, dedication, devotion, empathy, forbearance, forgiveness, friendship, generosity, gentleness, humanness, interdependence, kindness, patience, patriotism, reverence, sacrifice, selflessness, service, sharing, sympathy, thoughtfulness, tolerance and trust 5. Values related to NON-VIOLENCE are: • (a)PSYCHOLOGICAL: Benevolence, compassion, concern for others, consideration, forbearance, forgiveness, manners, happiness, loyalty, morality, and universal love • (b) SOCIAL: Appreciation of other cultures and religions, brotherhood, care of environment, citizenship, equality, harmlessness, national awareness, perseverance, respect for property, and social justice.	
(b)	Explain the role of Co-operation and commitment in ethical practice.	(7)
	COOPERATION It is a team-spirit present with every individual engaged in engineering. Co-operation is activity between two persons or sectors that aims at integration of operations (synergy), while not sacrificing the autonomy of either party. Further, working together ensures, coherence, i.e., blending of different skills required, towards common goals. Willingness to understand others, think and act together and putting this into practice, is cooperation. Cooperation promotes collinearity, coherence (blend), co-ordination (activities linked in sequence or priority) and the synergy (maximizing the output, by reinforcement).	

	The whole is more than the sum of the individuals. It helps in minimizing the input resources (including time) and maximizes the outputs, which include quantity, quality, effectiveness, and efficiency. According to professional ethics, cooperation should exist or be developed, and maintained, at several levels; between the employers and employees, between the superiors and subordinates, among the colleagues, between the producers and the suppliers (spare parts), and between the organisation and its customers. The impediments to successful cooperation are: 1. Clash of ego of individuals. 2. Lack of leadership and motivation. 3. Conflicts of interests, based on region, religion, language, and caste. 4. Ignorance and lack of interest. By careful planning, motivation, leadership, fostering and rewarding team work, professionalism and humanism beyond the 'divides', training on appreciation to different cultures, mutual understanding 'cooperation' can be developed and also sustained.	
13(a)	**Module 2** Explain the various reasons for an employee to behave unethically in an organization	(7)
	The researchers describe the different factors as "bad apples" (individual factors), "bad cases" (issue-specific factors) and "bad barrels" (environmental factors). Here's how they play out. 1. **Bad apples (individual factors)**: Unethical choices are more likely from people with specific personal characteristics — specific views and values. Overwhelmingly, these employees are driven by self-interest. For example, they manipulate others for their own personal gain, fail to see the connection between their actions and outcomes, and believe that ethical choices are driven by circumstance. They obey authority figures' unethical directives and act merely to avoid punishment. 2. **Bad cases (issue-specific factors)**: An employee might make an unethical choice in one situation, but not in others. Some issues are more likely to lead to unethical choices. Employees are more likely to act unethically when they don't see their action clearly causing harm — for example, when the victim is far away or the damage is delayed. Unethical choices also occur when an	

	employee feels that peers will not condemn their actions. 3. **Bad barrels (environmental factors):** Unethical choices are more likely when the organization encourages individualistic behaviour rather than doing what is best for other employees, customers, and the community. For example, the performance management system might reward individual bottom-line achievement, no matter how it is achieved. **Two things that don't influence ethical choices**: Research doesn't support two common ideas about what drives ethical behaviour. Demographic qualities —employee age, gender, and educational level — have little effect on unethical action. And, the existence of a code of conduct does not curb unethical actions, although enforcement of such a code does. The researchers suggest that stated codes of conduct have become so common that they have lost their power.	
(b)	What are the logical steps in solving moral dilemma?	(7)
	The logical steps in confronting moral dilemma are: 1. Identification of the moral factors and reasons. The clarity to identify the relevant moral values from among duties, rights, goods and obligations is obtained (conceptual inquiry). The most useful resource in identifying dilemmas in engineering is the professional codes of ethics, as interpreted by the professional experience. Another resource is talking with colleagues who can focus or narrow down the choice of values. 2. Collection of all information, data, and facts (factual inquiry) relevant to the situation. 3. Rank the moral options i.e., priority in application through value system, and also as obligatory, all right, acceptable, not acceptable, damaging, and most damaging etc. For example, in fulfilling responsibility, the codes give prime importance to public safety and protection of the environment, as compared to the individuals or the employers (conceptual inquiry). 4. Generate alternate courses of action to resolve the dilemma. Write down the main options and sub-options as a matrix or decision tree to ensure that all options are included 5. Discuss with colleagues and obtain their perspectives, priorities, and suggestions on various alternatives.	

6. Decide upon a final course of action, based on priority fixed or assumed. If there is no ideal solution, we arrive at a partially satisfactory or 'satisficing' solution.

14(a) Compare Gilligan's theory with Kohlberg theory on moral development. (7)

The theories of moral development by Kohlberg and Gilligan differ in the following respects.

Kohlberg's Theory	***Carol Gilligan's Theory***
A. Basic Aspects	
1. Is based on the study on men.	1. Is based on the study on men and women
2. Men give importance to moral rule.	2. Women always want to keep up the personal relationships with all the persons involved in the situations.
3. Ethics of rules and rights.	3. Women give attention to circumstances leading to critical situations rather than rules: (context-oriented and ethics of care)
B Characteristic Features	
1. Justice	1. Reason
2. Factual	2. Emotional
3. Right or wrong	3. Impact on relationships
4. Logic only	4. Compassion too
5. Logic and rule-based	5. Caring and concern
6. Less of caring	6. More of caring
7. Matter of fact (practical)	7. Abstract
8. Present focus	8. Future focus
9. Strict rules	9. Making exceptions
10. Independence	10. Dependence
11. Rigid	11. Human-oriented
12. Taking a commanding role	12. Shying away from decision-making
13. Transactional approach	13. Transformational approach

(b) Explain the term consensus and controversy in Engineering ethics. (7)

CONSENSUS AND CONTROVERSY

Consensus refers to agreement and controversy refers to disagreement. While exercising moral autonomy, individuals may differ in their decisions. Each may have different perspectives about morality and particular moral issues. There may be occasions in which the moral reasoning of one person will be in conflict with that of the other. During such occasions, each will resort to his moral autonomy which leads to controversy. Here, to avoid conflicts, everyone should discuss matter and reach a consensus.

15(a) **MODULE 3:** (7)

	Explain the role of 'Codes of Ethics' in the service life of a professional Engineer.	
	Engineers uphold and advance the integrity, honor and dignity of the engineering profession by: I. using their knowledge and skill for the enhancement of human welfare; II. Being honest and impartial, and servicing with fidelity the public, their employers and clients; III. Striving to increase the competence and prestige of the engineering profession; and IV. Supporting the professional and technical societies of their disciplines. A code of ethics prescribes how professionals are to pursue their common ideal so that each may do the best at a minimal cost to oneself and those they care about. The code is to protect each professional from certain pressures (for example, the pressure to cut corners to save money) by making it reasonably likely (and more likely then otherwise) that most other members of the profession will not take advantage. A code is a solution to a coordination problem. A professional has obligations to the employer, to customers, to other professionals- colleagues with specific expectations of reciprocity	
(b)	Explain the moral, conceptual, and factual issues that lead to challenger tragedy of 1986.	(7)
	Moral/Normative Issues: 1. The crew had no escape mechanism. Douglas, the engineer, designed an abort module to allow the separation of the orbiter, triggered by a field-joint leak. But such a 'safe exit' was rejected as too expensive, and because of an accompanying reduction in payload. 2. The crew were not informed of the problems existing in the field joints. The principle of informed consent was not followed. 3. Engineers gave warning signals on safety. But the management group prevailed over and ignored the warning. **Conceptual Issues**: 1. NASA counted that the probability of failure of the craft was one in one lakh launches. But it was expected that only the 100000th launch will fail. 2. There were 700 criticality-1 items, which included the field joints. A failure in any one of them would have caused the tragedy. No back-up or stand-bye had been provided for these criticality-1 components.	

	Factual/Descriptive Issues: 1. Field joints gave way in earlier flights. But the authorities felt the risk is not high. 2. NASA has disregarded warnings about the bad weather, at the time of launch, because they wanted to complete the project, prove their supremacy, get the funding from Government continued and get an applaud from the President of USA. 3. The inability of the Rockwell Engineers (manufacturer) to prove that the lift-off was unsafe. This was interpreted by the NASA, as an approval by Rockwell to launch.	
16(a)	Evaluate the importance of accountability in a professional's life.	(7)
	Accountability means: 1. The capacity to understand and act on moral reasons 2. Willingness to submit one's actions to moral scrutiny and be responsive to the assessment of others. It includes being answerable for meeting specific obligations, i.e., liable to justify (or give reasonable excuses) the decisions, actions or means, and outcomes (sometimes unexpected), when required by the stakeholders or by law. 3. Conscientiousness: It means: (a) Being sensitive to full range of moral values and responsibilities and (b) The willingness to upgrade their skills, put efforts, and reach the best balance possible among those considerations, and 4. Blameworthy/Praiseworthy: Own the responsibility for the good or wrong outcomes. Courage to accept the mistakes will ensure success in the efforts in future. Although the engineers facilitate experiments, they are not alone in the field. Their responsibility is shared with the organizations, people, government, and others. No doubt the engineers share a greater responsibility while monitoring the projects, identifying the risks, and informing the clients and the public with facts. Based on this, they can take decisions to participate or protest or promote.	
(b)	Evaluate how an Engineer can be a responsible experimenter.	(7)
	The engineer, as an experimenter, owe several responsibilities to the society, namely, 1. A conscientious commitment to live by moral values. 2. A comprehensive perspective on relevant information. It includes constant awareness of the progress of the experiment and readiness to monitor the side effects, if any. 3. Unrestricted free-personal involvement in all steps of the project/product development (autonomy).	

	4. Be accountable for the results of the project (accountability).	
17(a)	**MODULE 4** Explain the various justifications for confidentiality.	(7)
	Confidentiality means keeping the information on the employer and clients, as secrets. It is one of the important aspects of team work. Confidentiality can be justified by various ethical theories. According to Rights-based theory, rights of the stakeholders, right to the intellectual property of the company are protected by this practice. Based on Duty theory, employees and employers have duty to keep up mutual trust. The Utilitarian theory holds good, only when confidentiality produce best to most people. Act utilitarian theory focuses on each situation, when the employer decides on some matters as confidential. 1. Respect for Autonomy: It means respecting the freedom and self-determination of individuals and organizations to identify their legitimate control over the personal information of themselves. In the absence of this, they can not keep their privacy and protect their self-interest 2. Respect for Promises: (Self-explanatory) 3. Trustworthiness: (Self-explanatory) 4. Respect for public welfare: (Self-explanatory)	
(b)	Explain how you can improve collegiality in an organization where you are presently employed.	(7)
	Collegiality is the term that describes a work environment where responsibility and authority are shared among the colleagues. When Engineering codes of ethics mention collegiality, they generally cite acts that constitute disloyalty. The disloyalty of professionals towards an organization, reflects the attitude they have towards the work environment for the salaries they are paid and the trust the company has for them. Collegiality is the relationship between colleagues. Colleague is taken to mean a fellow member of the same profession, a group of colleagues united in a common purpose, and used in proper names, such as Electoral College, College of Cardinals, and College of Pontiffs. Colleagues are those explicitly united in a common purpose and respecting each other's abilities to work toward that purpose. A colleague is an associate in a profession or in a civil or ecclesiastical office. Collegiality can connote respect for another's commitment to the common purpose and ability to work toward it. In a narrower sense, members of the faculty of a university or college are each other's colleagues. Sociologists of organizations use the word collegiality in a technical sense, to create a	

	contrast with the concept of bureaucracy. Classical authors such as Max Weber consider collegiality as an organizational device used by autocrats to prevent experts and professionals from challenging monocratic and sometimes arbitrary powers. The main factors that help in maintain harmony among members at a workplace are: · Respect · Commitment · Connectedness	
18(a)	Explain the significance of different types of Authority in an organization.	(7)
	Institutional Authority: It is the authority exercised within the organization. It is the right given to the employees to exercise power, to complete the task and force them to achieve their goals. Duties such as resource allocation, policy dissemination, recommendation, supervision, issue orders (empower) or directions on subordinates are vested to institutional authority, e.g., Line Managers and Project Managers have the institutional duty to make sure that the products/projects are completed successfully. The characteristics features of institutional authority are that they allocate money and other resources and have liberty in execution. **Expert Authority**: On the other hand, the Expert Authority is (a) the possession of special knowledge, skills and competence to perform a job thoroughly (expertise), (b) the advice on jobs, and (c) is a staff function. It is also known as 'authority of leadership'. These experts direct others in effective manner, e.g., advisers, experts, and consultants are engaged in an organization for a specific term.	
(b)	Discuss about the various rights of an engineer.	(7)
	Right of Professional Conscience: This is a basic right which explains that the decisions taken while carrying on with the duty, where they are taken in moral and ethical manner, cannot be opposed. The right of professional conscience is the moral right to exercise professional judgement in pursuing professional responsibilities. It requires autonomous moral judgement in trying to uncover the most morally reasonable courses of action, and the correct courses of action are not always obvious. Right of Conscientious: Refusal The right of conscientious refusal is the right to refuse to engage in unethical behavior. This can be done solely because it feels unethical to the doer. This action might bring conflicts within the authority-based relationships.	

	Right to Recognition: An engineer has a right to the recognition of one's work and accomplishments. An engineer also has right to speak about the work one does by maintaining confidentiality and can receive external recognition. The right for internal recognition which includes patents, promotions, raises etc. along with a fair remuneration, are also a part of it.	
19(a)	**MODULE 5:** Explain human centred Environmental ethics with nature centred ethics.	(7)
	Environmental ethics is the study of (a) moral issues concerning the environment, and (b) moral perspectives, beliefs, or attitudes concerning those issues. Engineers in the past are known for their negligence of environment, in their activities. It has become important now that engineers design eco-friendly tools, machines, sustainable products, processes, and projects. These are essential now to (a) ensure protection (safety) of environment (b) prevent the degradation of environment, and (c) slow down the exploitation of the natural resources, so that the future generation can survive. Two World Views on Environmental Ethics: (a) Anthropocentric Worldview: This view is guiding most industrial societies. It puts human beings in the center giving them the highest status. Man is considered to be most capable for managing the planet earth. (b) Eco-centric Worldview: This is based on earth-wisdom. The basic beliefs are as follows: 1. Nature exists not for human beings alone, but for all the species. 2. The earth resources are limited and they do not belong only to human beings. 3. Economic growth is good till it encourages earth-sustaining development and discourages earth-degrading development. 4. A healthy economy depends upon a healthy environment. 5. The success of mankind depends upon how best we can cooperate with the rest of the nature while trying to use the resources of nature for our benefit.	
(b)	Explain the different types of issues in computer ethics.	(7)
	Different types of problems are found in computer ethics. 1. Computer as the Instrument of Unethical Acts:	

	(a) The usage of computer replaces the job positions. This has been overcome to a large extent by readjusting work assignments, and training everyone on computer applications such as word processing, editing, and graphics. (b) Breaking privacy. Information or data of the individuals accessed or erased or the ownership changed. (c) Defraud a bank or a client, by accessing and withdrawing money from other's bank account. 2. Computer as the Object of Unethical Act: The data are accessed and deleted or changed. (a) Hacking: The software is stolen or information is accessed from other computers. This may cause financial loss to the business or violation of privacy rights of the individuals or business. In case of defense information being hacked, this may endanger the security of the nation. (b) Spreading virus: Through mail or otherwise, other computers are accessed and the files are erased or contents changed altogether. 'Trojan horses' are implanted to distort the messages and files beyond recovery. This again causes financial loss or mental torture to the individuals. Some hackers feel that they have justified their right of free information or they do it for fun. However, these acts are certainly unethical. (c) Health hazard: The computers pose threat during their use as well as during disposal. 3. Problems Related to the Autonomous Nature of Computer (a) Security risk (b) Loss of human lives (c) In flexible manufacturing systems, the autonomous computer is beneficial in obtaining continuous monitoring and automatic control.	
20(a)	Discuss about the role of engineers as expert witness.	(7)
	Frequently engineers are required to act as consultants and provide expert opinion and views in many legal cases of the past events. They are required to explain the causes of accidents, malfunctions and other technological behavior of structures, machines, and instruments, e.g., personal injury while using an instrument, defective product, traffic accident, structure or building collapse, and damage to the property, are some of the cases where testimonies are needed. The focus is on the past. The functions of eye-witness and expert-witness are different as presented in the Table below	

Eye-witness	*Expert-witness*
1. Eye-witness gives evidence on only what has been seen or heard actually (perceived facts)	1. Gives expert view on the facts in their area of their expertise 2. Interprets the facts, in term of the cause and effect relationship 3. Comments on the view of the opposite side 4. Reports on the professional standards, especially on the precautions when the product is made or the service is provided

(b)	What are the various conflict situations faced by a project manager managing a work site?	(7)

(a) Conflicts based on schedules: This happens because of various levels of execution, priority and limitations of each level.

(b) Conflicts arising out of fixing the priority to different projects or departments. This is to be arrived at from the end requirements and it may change from time to time.

(c) Conflict based on the availability of personnel.

(d) Conflict over technical, economic, and time factors such as cost, time, and performance level.

(e) Conflict arising in administration such as authority, responsibility, accountability, and logistics required.

(f) Conflicts of personality, human psychology and ego problems.

(g) Conflict over expenditure and its deviations.

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15(a)

(b) Explain the moral, conceptual, and factual issues that lead to challenger tragedy of 1986.

16(a) Evaluate the importance of accountability in a professional's life. (7)

(b) Evaluate how an Engineer can be a responsible experimenter. (7)

Module 4

17(a) Explain the various justifications for confidentiality. (7)

(b) Explain how you can improve collegiality in an organisation where you are presently employed. (7)

18(a) Explain the significance of different types of Authority in an organisation. (7)

(b) Discuss about the various rights of an engineer. (7)

Module 5

19(a) Explain human centred Environmental ethics with nature centred ethics. (7)

(b) Explain the different types of issues in computer ethics. (7)

20(a) Discuss about the role of engineers as expert witness. (7)

(b) What are the various conflict situations faced by a project manager managing a work site? (7)

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